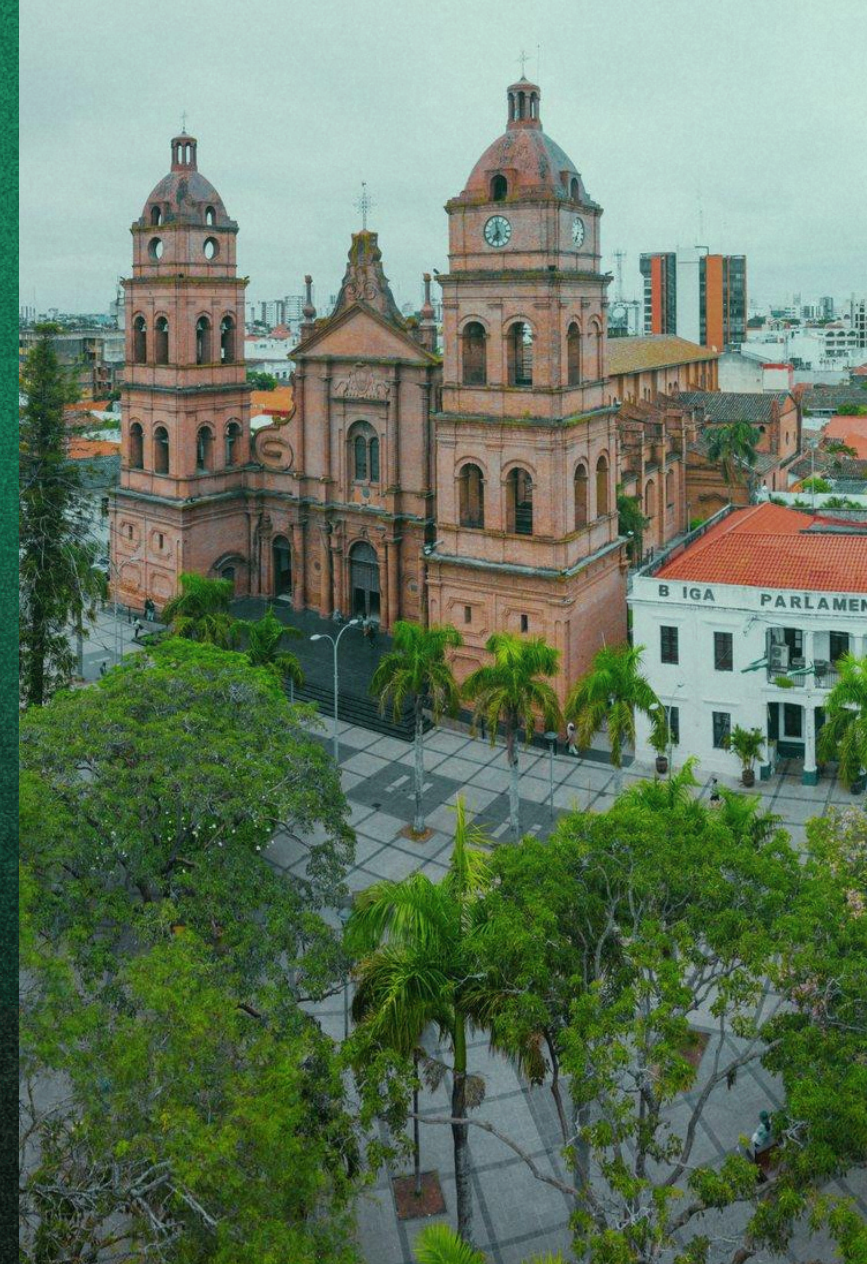


SANTA CRUZ, BOLIVIA | 29 DE AGOSTO DE
2026

aws
COMMUNITY DAY
— **BOLIVIA** —



ROAD TO AWS **COMMUNITY DAY** BOLIVIA

Terraform at Scale: Security Without Friction

Amaury Borges Souza

Senior Cloud Security Engineer at CI&T

MBA Professor at FIAP

AWS Community Builder

Hashicorp Champion

Prof. Amaury Borges Souza

Cloud Security Engineer @CI&T



Amaury Borges Souza

He/Him/His
Senior Cloud Security Engineer
CI&T

Security-focused Cloud Engineer with over 12 years of experience in the field. I also enjoy writing. I am a recognized community leader (AWS & HashiCorp). I have an interest in AI/ML and a passion for automation tools, ethical hacking, and Linux.

- Professor at FIAP (Faculty member for the DevOps, Automation & Cloud Strategy MBA program)
- **AWS Community Builder** (Supporting the AWS cloud and services community)
- HashiCorp Ambassador (Fan of Terraform, IaC, automation, and DevSecOps)
- HashiCorp User Group Leader (Leading the HashiCorp community in Campinas)



AWS Community Builder

4x AWS Community Builder

Sharing knowledge through talks, articles & hands-on content.

Connecting Cloud Security, AWS & AI Technologies.

Contributing to the community through events and education.



Tables of Contents for AWS Community Day

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| 01 | Cloud Security Reality | 05 | AWS Security Agent |
| 02 | AWS Guardrails | 06 | Policy as Code |
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| 04 | Secure Remote on AWS | | |

Cloud Security: The Reality

29%

of cyber incidents involved cloud environments

Unit 42 found cloud environments involved in nearly 29% of the incidents analyzed; 21% caused an operational impact on cloud assets or environments.

53%

report permissive IAM as a top cloud security challenge

The State of Cloud Security 2025 identifies permissive IAM practices as one of the key challenges and exfiltration vectors.

22%

of breaches started with compromised credentials

This data point from the 2025 Verizon DBIR ties in perfectly with your slide on OIDC vs. long-lived AWS access keys.

IaC Usage is Crucial to Provisioning Infrastructure

Infrastructure as Code Market

The use of Infrastructure as Code (IaC) has become a de facto standard; let's explore the drivers, common approaches, and trends.

IaC is crucial for software-defined infrastructure

100
75
50

90%
of cloud users surveyed are using IaC

DevOps pipelines are the preferred deployment method

Nearly 60% use their existing DevOps pipelines to deploy IaC. Dedicated IaC tools such as Env0, Spacelift, Scalr or Atlantis are nascent at less than 20% use while Terraform Cloud/Enterprise is used by roughly half of the respondents.

Adding dedicated CI/CD for IaC can add to the complexity and knowledge requirements so it is not surprising that many adhere to familiar CI/CD tools used for application software.

DevOps pipeline: Jenkins, Github Actions, Gitlab CI, or Other	58%
Terraform Cloud/Enterprise	53%
Running Terraform locally	32%
Other	21%
Env0, Spacelift, Scalr or Atlantis	19%

MERCADO DE IAC (TRENDS)

O uso de infraestrutura como código (IaC) se tornou um padrão de fato, vamos entender os motivadores, abordagens comuns e tendências.

The top challenges to managing cloud infrastructure

Security (41%)

Visibility (41%)

Knowledge (40%)

Governance (36%)

Leaked AWS keys give full control over corporate accounts

Tracking this exposure for the past four years
More than 9,300 Amazon Web Services (AWS) access keys publicly exposed between August 2022 and August 2026 are still active and valid.

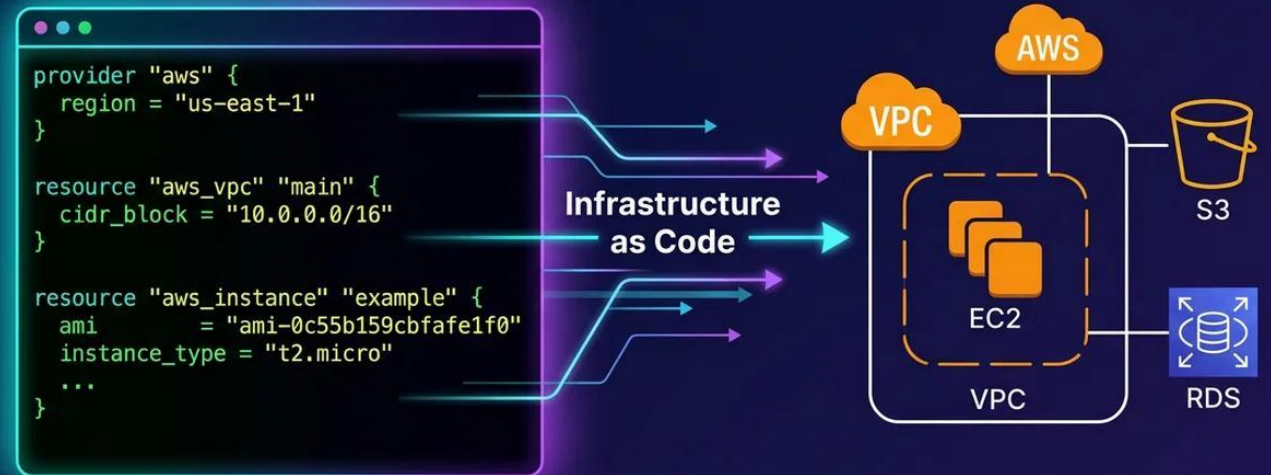


Terraform Changed How We Build Cloud

From manual infrastructure to infrastructure at scale

- Infrastructure became code.
- Deployments became repeatable.
- Teams gained autonomy.
- Cloud environments started scaling faster.

Terraform × AWS



Terraform didn't just automate infrastructure. It changed how infrastructure scales.

Dinamic Questions

- Who here has ever opened a Security Group to 0.0.0.0/0 just to test something?
- What causes more incidents today: vulnerabilities or misconfigurations?
- Who has used Terraform? Checkov? CloudFormation?

“

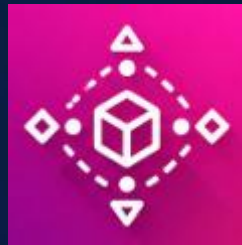
Here, we need to reinforce the security **AWS-native** foundation before any third-party tools. It's the protect the environment against provisioning.

Structural Guardrails in AWS



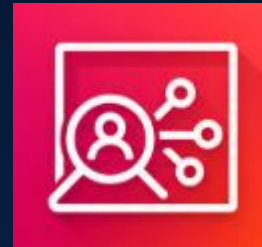
AWS Organizations + SCPs

Deny actions at the org level, before Terraform ever runs (e.g. block public S3 buckets even if someone tries)



Control Tower / Landing Zone

Every new account is born with guardrails already in place



IAM Access Analyzer

Surfaces unintended cross-account and external access



Key takeaway

Your first line of defense isn't the IaC scanner, it's your organization's architecture.

Secure Infrastructure Delivery with Terraform

Security Infrastructure as code

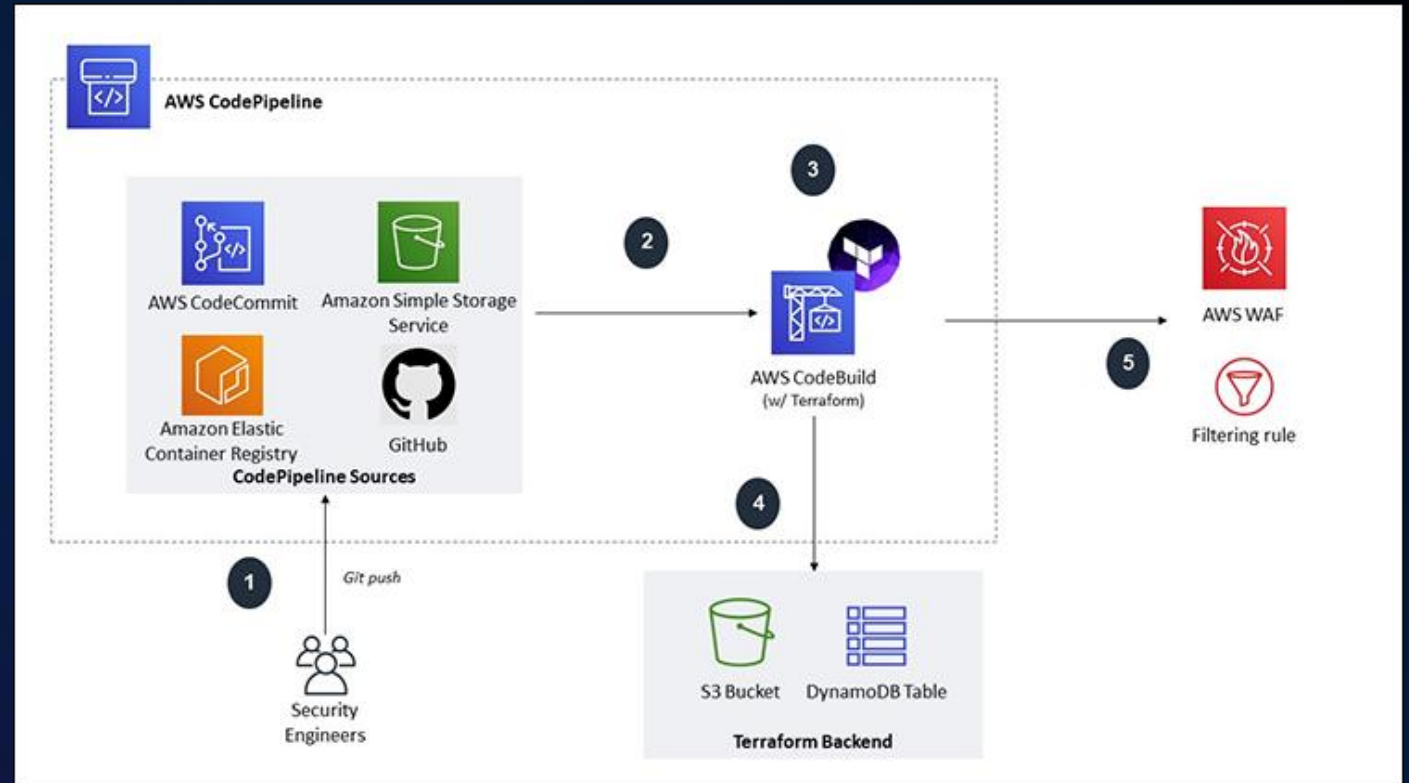
Define security controls as versioned and repeatable Terraform code.

Automated deployments through CI/CD

Changes are reviewed, validated and deployed through an automated pipeline.

Terraform executed by AWS CodeBuild

CodeBuild runs Terraform to provision and update AWS resources consistently.



abiydv.github.io

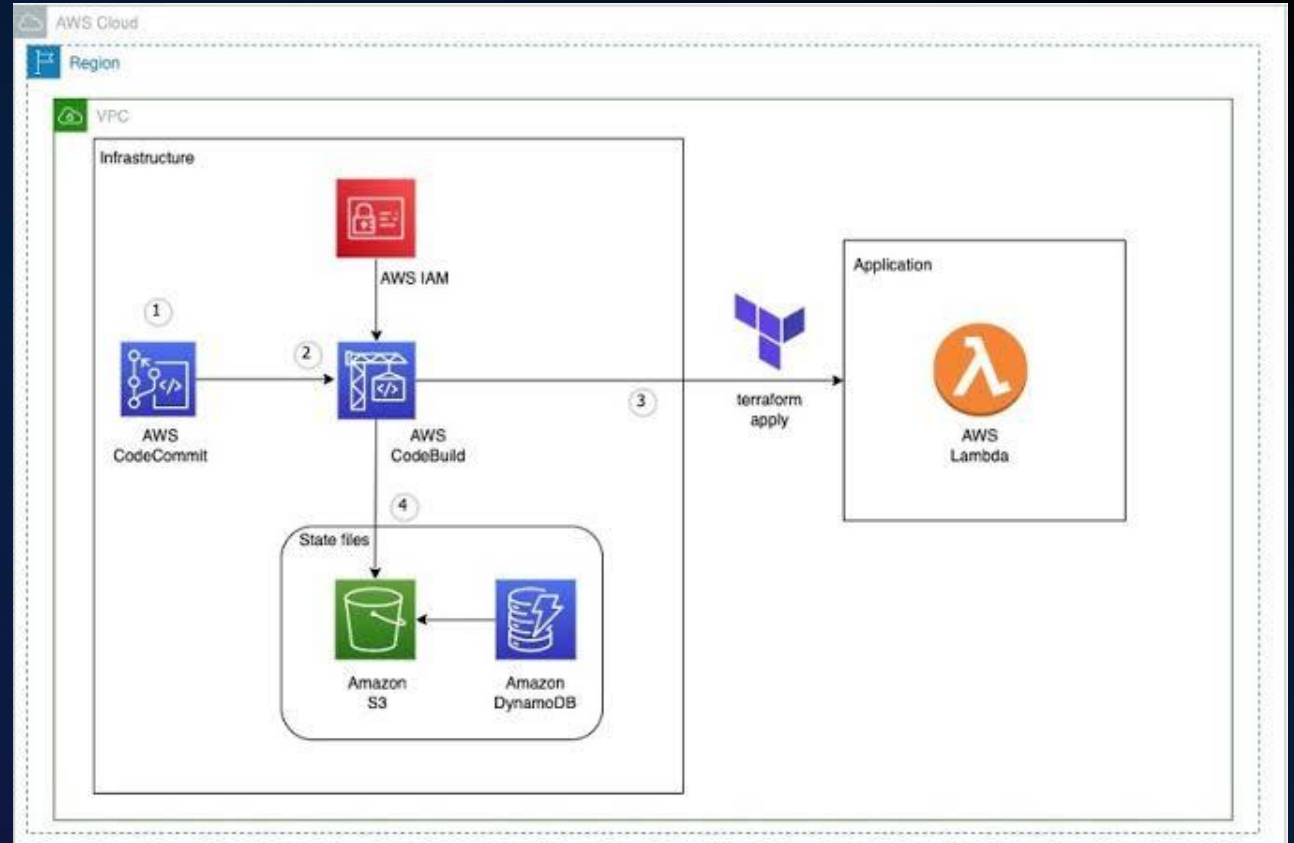
3 Pillars of Secure Terraform at Scale

Secure Remote Terraform State on AWS

Protecting sensitive data and Terraform state at scale.

- Centralized remote state
- S3 versioning for recovery
- Encryption at rest
- Least-privilege IAM access
- Native S3 state locking
- Separate state per environment

Locking can be enabled via **S3** or DynamoDB. However, DynamoDB-based locking is **deprecated**



Secure AWS Authentication for Terraform

No static
AWS access
keys

Short-lived
credentials

IAM roles with least
privilege

Automatic
credential
rotation

Secure Terraform
code

IAM/OIDC



Multi-Account by Design

AWS Account Isolation

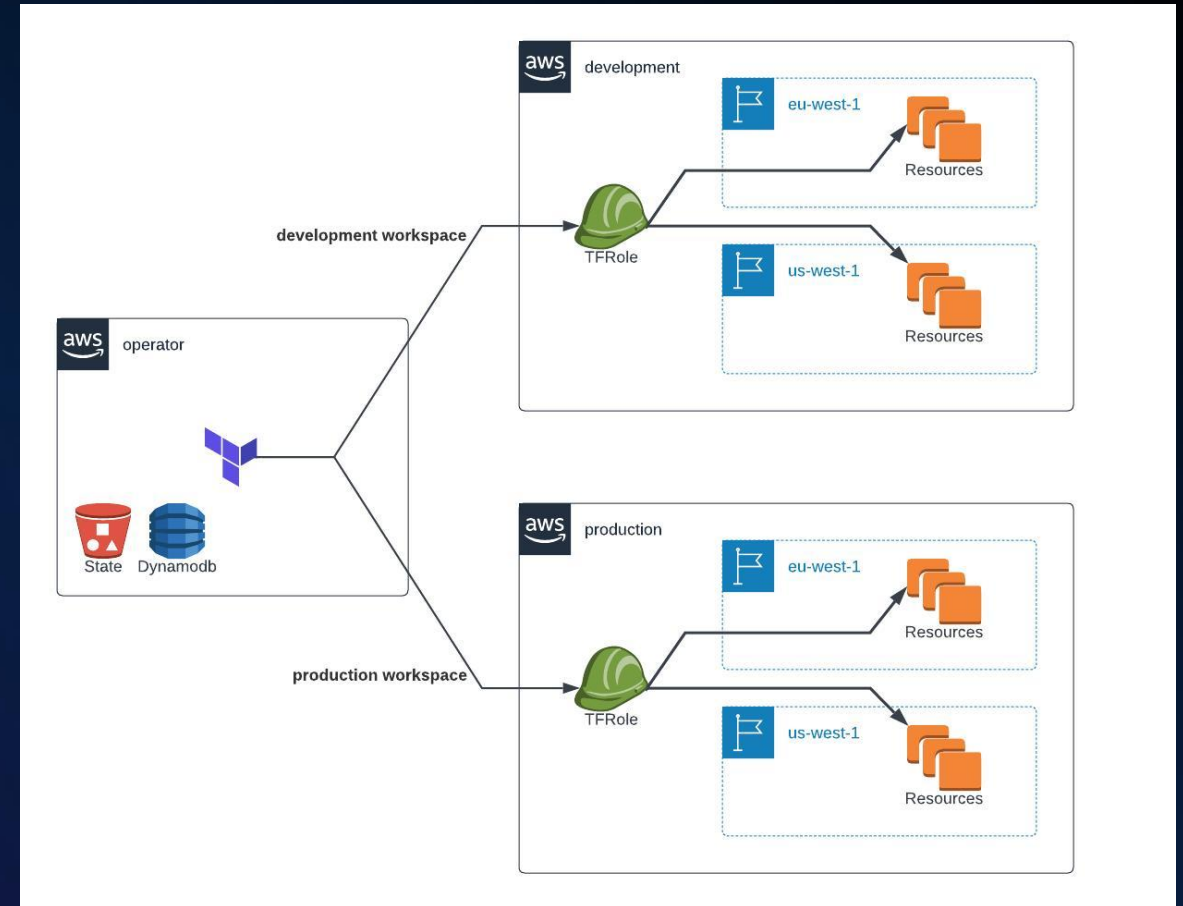
Separate dev, staging and production blast radius.

Independent Terraform State

One state boundary per environment/account.

Governance at Organization Level

AWS Organizations + SCPs provide guardrails above Terraform.



abiydv.github.io

How KIRO and AWS Security Agent can help

Agentic Infrastructure

AI AGENTS ARE CHANGING HOW WE BUILD AND OPERATE INFRASTRUCTURE AS CODE.

Introducing Kiro CLI



```
○ kiro × + ▾  
  
KIRO  
  
Welcome to the new Kiro CLI UX! Use /feedback to share your thoughts.  
Prefer the classic experience? Relaunch with kiro-cli --classic.  
  
Can you build me a cool snake game using subagents to build it?  
  
Let me build you a snake game using subagents! I'll split the work into parallel tracks for design/planning and then implementation.  
○ Orchestrating (3 agents)  
  ○ implementer kiro_default ⚠tool approval needed  
  ● designer kiro_default Completed  
  
⚠1 tool approval pending from subagents  
> (a) Approve all pending  
  (f) Approve all pending and auto-approve all future requests  
  (c) Configure individually (agent monitor)  
  (x) Exit (cancel subagents)
```


From IaC Scanning to Agentic Security

AI AGENTS ARE CHANGING HOW WE BUILD AND OPERATE INFRASTRUCTURE AS CODE.

Context-aware security reviews

Understands code, architecture and requirements

Threat Modeling

Identifies threats and attack paths

Code Review

Security analysis across repositories

Agentic Pentesting

Discovers and validates exploitable vulnerabilities

AWS Security Agent

Home > demo > Code reviews > catsdogs-review > June 7, 2026 code review run

June 7, 2026, 17:28 (UTC+9:00) Complete

Code review: catsdogs-review Last update: June 7, 2026, 18:47 (UTC+9:00)

Generate report Stop run

Completed Preflight Completed Static analysis Completed Validation Completed Finalizing

Code review run Preflight Code review logs Findings

Findings (12) Showing all findings

Remediate code

Filter findings

EC2 instance metadata service v1 enabled by default allows credential theft via SSRF

Confidence - High High Confirmed

Last updated: June 7, 2026, 18:36 (UTC+9:00)

DOM XSS via unescaped search term and indexed content in Hugo theme search

Low Unconfirmed

Finding

EC2 instance metadata service v1 enabled by default allows credential theft via SSRF

Last updated: 6/7/2026, 6:36:53 PM · Version 6

Status Active

Agent confidence Confidence - High

Severity High

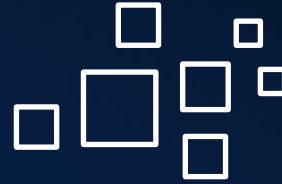
Validation Status Confirmed

Risk Score -

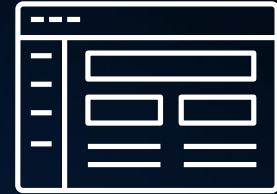
5 Tools for Secure IaC



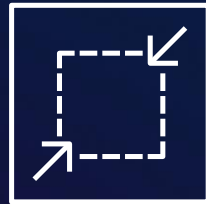
**AWS CloudFormation
Guard**



Checkov



OPA/Conftest



TFPolicy



AWS Security Hub CSPM



“

**More automation requires
stronger guardrails, not
more manual approvals.**

AWS Community Meetup Speaker
AWS U.G. Campinas

Policy as Code: in the pipeline

Fully-featured policy-as-code

Static IaC
Analysis

Built-in
Policies

Custom Policies

Terraform Plan
Scanning

Tienen imágenes
sencillas

Tienen un mensaje
clave por diapositiva

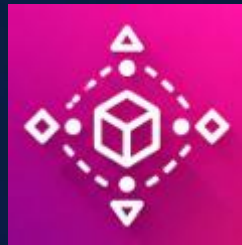
```
metadata:
  id: "CKV2_AWS_17"
  name: "Ensure that EC2 instances belong to a VPC"
  category: "NETWORKING"
definition:
  and:
    - resource_types:
      - aws_instance
      connected_resource_types:
      - aws_network_interface
      operator: exists
      cond_type: connection
    - resource_types:
      - aws_network_interface
      connected_resource_types:
      - aws_subnet
      operator: exists
      cond_type: connection
    - resource_types:
      - aws_subnet
      operator: exists
      attribute: vpc_id
      cond_type: attribute
    - cond_type: filter
      attribute: resource_type
      value:
      - aws_network_interface
      operator: within
```

Policy as Code Tools



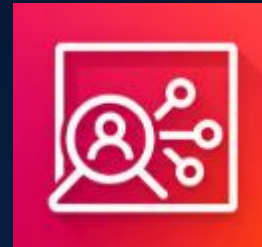
Checkov

Deny actions at the org level, before Terraform ever runs (e.g. block public S3 buckets even if someone tries)



Hashicorp Sentinel

Every new account is born with guardrails already in place



IAM Access Analyzer

Surfaces unintended cross-account and external access



Key takeaway

Your first line of defense isn't the IaC scanner, it's your organization's architecture.

Avoid

Long-lived AWS access keys

Local or shared state without controls

Security only after terraform apply

Manual security reviews for every change

Prefer

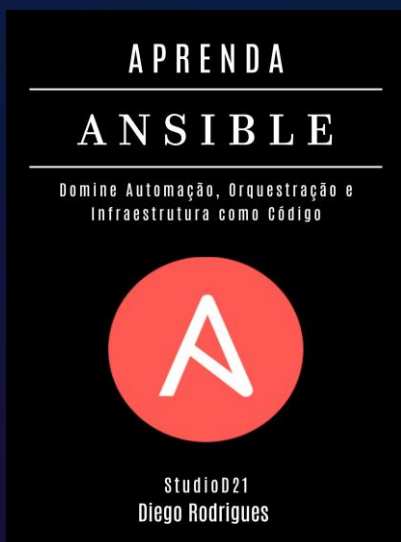
OIDC + temporary IAM roles

Remote S3 state + encryption + locking

IaC scanning before deployment

Policy as Code + automated guardrails

Additional Resources: Books



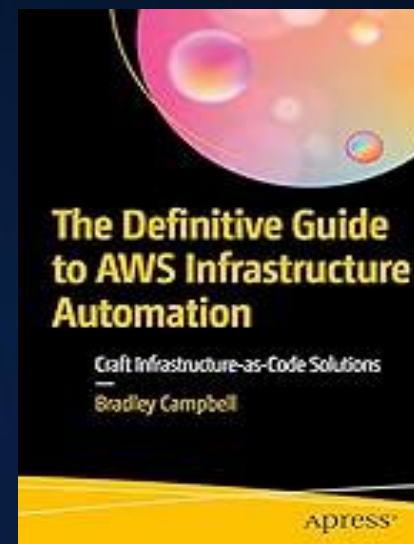
Learn Ansible

Diego Rodrigues (describes patterns and practices for building and developing infrastructure as code).



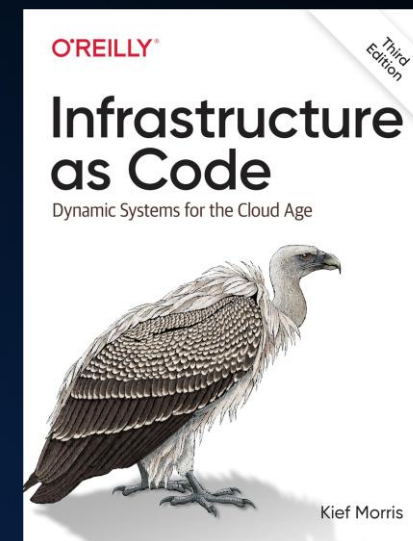
Terraform Up & Running

Yevgeniy (A practical roadmap for improving IT in any organization).



The Definitive Guide to AWS Infrastructure

Bradley (explores how DevOps and security techniques should be applied together).



Infrastructure as Code

Kief Morris (This book is an engaging read that brilliantly captures the dilemmas faced by companies).

“

I appreciate the AWS Community Day Bolivia Team, and AWS User Group Cochabamba, for CFP accepted and invite me to talk here.

Amaury Borges Souza

¡Gracias!

Amaury Borges Souza

Senior Cloud Security Engineer

FIAP Professor | AWS Community Builder

Terraform & Cloud Security



Complete la encuesta de la sesión en el código QR